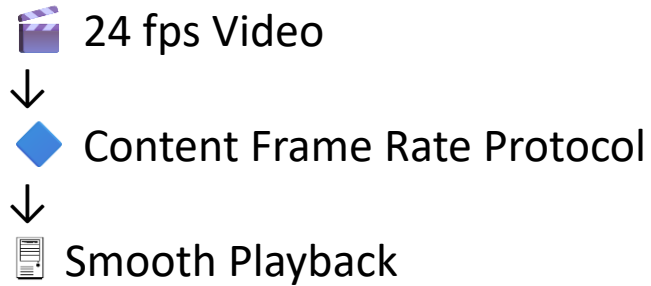


Introducing the Wayland Content Frame Rate Protocol

Frame-Rate-Aware Presentation for Modern Linux Desktops



XDC 2026 • LIGHTNING TALK (5 MINUTES)

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Why Video Playback Judders Today

24 fps Video
↓
Media Player
↓
FPS Metadata Lost
↓
Compositor
↓
60 Hz Display

Compositor is blind to content frame rate

3:2 Pulldown


VRR 40-60Hz
24fps ✗
30fps ✗

How wp-content-frame-rate-v1 Fixes It

24 fps Video



Media Player

set_frame_rate()



wp-content-frame-rate-v1



Compositor



Refresh-Aware Scheduling

Fixed

24fps→24/48Hz

Perfect Cadence

VRR

24fps→LFC→48Hz

Inside Range

Before: 

After: 

Making Frame Rate a First-Class Citizen in Wayland

BEFORE

24fps→60Hz

3:2 cadence

Judder

wp-content-frame-rate-v1



AFTER

set_frame_rate()

24/48Hz

1:1 cadence

Smooth Playback

UPSTREAM STATUS: Ready for review and Integration

Wayland Protocol #518 • Mutter #5091 • KWin #9882 • Weston #2203 • MPV #18038

Thank you